



Interoffice Memo

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SUBJECT: Zero Point Energy, Defense Intelligence Agency Meeting December 1989, Washington, D.C.

KEY WORDS: Radar; Medical x-rays; Tesla coils; Cold fusion; Propulsion; Satellite launch agreements; Japanese - Soviet joint exploration of space.

SUMMARY

This was the 6th meeting held on scalar waves/zero point energy. United States interest is due to the resurgence of interest within the Soviet Union and their ability to apply the results of Tesla's experiments to this area. The whole area is considered a fundamental enabling technology. High resolution, flat displays exhibiting no degradation with age and able to be seen in high light situations, is a result of the application of condensed charged technology which is nothing more than another name for zero point energy. Radar systems above 90 GHz are possible using plastic wave guides and resistant to electromagnetic propagation. The mechanism of charge transfer is not limited by charge carrier density which means perhaps a hundred fold increase in the speed of present day computers. A new algebra referred to as "Clifford" algebra is necessary to calculate what is referred to as the 'A fields' and polarization modulation. The gauge group is SU-2, which means the charges do not repel. The 'A potential' is a gauge field and has not affected any of our present day electrical engineering. It is not a mathematical convenience, as is the choice of Heaviside vector notation to describe Maxwell's equations. Verifying the existence of 'A vector fields' (a curl-free vector potential) is difficult. An experiment using a toroidal coil shielded against the 'B fields' was described. A Josephson detector has

been designed to verify the existence of the 'A field'. The application of zero point energy in scalar fields to devices employing cathodes means the cathodes last almost 20 years. The so-called "Kraysnoiarsk Radar", the subject of recent concern to the United States, is not a radar but is a scalar transmitter to transfer SU-2 power to satellites and then on to Moscow. We in this country tend to look at laser technology; the Soviets are more involved with electromagnetic radiation effects. There is an error in the information originally transmitted describing input power and output power. The input power is 700 watts, not 100 watts, as originally described. The output power is not 3,000 kilowatts, but rather 3,000 watts. The power levels are still impressive, regardless of the misinformation originally given. The tube involved in the generation of this power was originally described by Tesla and used by him in his Colorado Springs experiments. An agreement was also described between the Soviets and Japan, where the Soviets would furnish the heavy launch capability to place a joint Soviet-Japanese space station in low earth orbit, with subsequent bases on the moon and the journey to Mars. The Soviets would provide the heavy lift capability and the power system based on zero point energy for the joint space station. The Japanese will supply all the technology for the moon base, the space station, and the trip to Mars program. The feeling is the Soviets want to tie all of Europe together. The Soviets will get manufacturing technology with everyone else involved getting technology from the Soviet Union. DIA projects the French space company, Ariane, will tie up 100 commercial launches over the next decade. They presently have 75 launches. The concern on the part of the DIA is this leaves 15 military launches for the United States. The general consensus of opinion of the attendees at the meeting was that due to our being too security conscious regarding our space launch complexes, it simply takes us too long to launch a commercial satellite.

DETAILS:

BACKGROUND TO MEETINGS. This meeting was the 6th meeting in a series. The other meeting dates were ■ February 1988; ■ July 1988; November 1988; March 1989; and June 1989. Dr. ■ of the DIA transferred all responsibility for future meetings to Dr. ■ will be retiring in December, but will be retained by the DIA to advise ■ on this area. The Intelligence Research and Development Council (IRDC), a government organization composed of DIA, Joint Chiefs of Staff and other organizations, has identified zero point energy/scalar wave technology as an "up and coming" area. The DIA would like to classify the entire area; the research community feels classification at this time is premature. The concept is based on Tesla's material presently in the hands of both the United States and the Yugoslaves; the latter having passed it on to the Soviet Union. Tesla felt there was energy to be tapped in the cosmos. Zero point energy is kinetic energy. The question which needs an answer is simply: Is zero point energy real or virtual? If it is virtual, then the energy will be difficult to tap into. Fields drive particle "giggles" and the "giggles" generate

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electromagnetic fields. The amount of energy of the cosmos, if it can be tapped, should be enormous. The Undersecretary of Defense has been briefed on this area. Because of the potential, the office of the Secretary of Defense reordered research priorities and placed zero point energy/scalar waves into second place. This would be a good time for Hercules to become involved in an up and coming technology, not necessarily related to either materials or propulsion, but which could form a fallback position, should either of our prominent strategies fail.

ZERO ENERGY/SCALAR WAVE APPLICATIONS. The whole area of zero point energy/scalar waves can be classified as condensed charge technology (CCT). The general feeling is CCT represents a fundamental enabling technology. Several organizations are doing the theoretical work on this area. These studies indicate radars can be developed having frequencies greater than or equal to 90 GHz, energy storage devices very similar to capacitors or superconductivity; medical devices, high frequency communication techniques which are virtually secure without any further encryption, and electromagnetic propagation-free devices. Present limitations to electronic devices result because charge carrier densities build up and limit the speed of these devices. Both high charge and high power densities can be obtained from electron droplet condensed charge beads - 10^{12} electrons in about 1 micron diameter (called "E" beads). If more electrons are involved, then a beaded chain is obtained containing 10^{12} electrons; 100 beads to the chain contain 10^{15} electrons. The physical evidence for these chains is found in electron beams, which arrive in spots of about 1 micron in diameter. Low voltage vacuum also show cathode spot phenomenon (Laferty's book entitled "Vacuum Arcs" illustrates many of these phenomenon). Similar results are obtained in air as in vacuum. The charge arrives in discreet bundles. Any extra beads tend to hit in the center of the ring. The charge density determines the ring size. These discreet bundles are also seen in cosmic effects where elementary particles are made from quarks. The feeling was also expressed that the Casimier effect generates zero point energy. The Casimier effect is a driving force which forces two plates of a capacitor together with tremendous force over extremely short distances which are on the order of microns. These forces can also be seen as quantum noise in an oscillator. The universe as a whole is a cavity with energy equivalent of 10^{94} grams/cm³, a value calculated for this effect.

Zero point energy is completely homogenous and isotropic. Spontaneous emission is caused by zero point energy and the Lamb shift of 1 GHz is readily explained from theories regarding zero point energy. Gravitational effects are considered residual effects from zero point energy, and therefore not a true force. This statement alone, if proven, will rewrite the physics books. The proverbial question "Why doesn't the electron blow itself up" can be answered using zero point energy concepts. The radiation pressure from zero point energy holds it together. Zero point energy is scalable and instead of being modeled as a sphere, should be modeled as a Taurus.

SOURCES OF ZERO POINT ENERGY. Zero point energy devices use a cathode and an anode at opposite ends of a tube. The signal is taken off the anode side with a load resistor to ground. Along one side of the tube (which can be glass or plastic), a plate or a half cylinder is placed and grounded. The cathode is a point of material resting against another plate which is wetted with mercury. In addition, a separator is employed to hold back ions so just the zero point energy flows through. Wave guides are manufactured by cutting a groove in a plastic and the charge follows the contour of the groove. Flat panel displays can be manufactured this way using very fine lines to enhance resolution. The charge literally flows through the cut grooves in the screen.

Electron beam machining apparatus is a good source of zero point energy. The Japanese, for instance, have used electron beam machining for turbo pump impellers on their liquid rocket engines. The French used a similar machining operation. The generation of the various ring structures created craters and in each crater a crack was formed which was nothing more than a stress riser. When the part was used several times, both for testing and flights, the part disintegrated. Low technology machining techniques do well however, because no craters are formed.

An x-ray device using zero point energy was also described. Very similar to the cathode device, described above, 10 KeV accelerated electrons yielded greater than 10 MeV radiation. However, an input of 3 KV yielded only 3 KeV x-rays. The EV velocity is 0.1 c; the charge is $10^{19}E$. The pulse length was 1.4 ma at 1.1 nano seconds (100 cycles) peak power radiated.

For flat panel displays, 20 micron channels give 8,000 lines per inch. DARPA presently is under contract to one of the companies for a device using zero point energy and flat panels. Bob Forward has proposed extracting energy from the vacuum using the Casimier effect. Basically, 2 plates are allowed to explosively come together due to zero point energy forcing them together. After the plates come together explosively and generate their heat, they are discarded. Another use would be as a traveling wave tube where the cathode and anode and the joining glass tube are surrounded by a helix. This is the type of a tube used by Tesla/Colorado Springs to generate an uncontrolled tapping of zero point energy. Regrettably, Tesla could not control the zero point energy he tapped into and it fed back through his apparatus into the central power station and caused the generators to catch fire.

THEORETICAL CONSIDERATIONS OF ZERO POINT ENERGY. The field caused by zero point energy is referred to as an "A field". The symmetry group is SU-2 which implies that charge does not repel. The A fields are Abelian groups, which means 1×2 equals 2×1 , which equals zero. (Non-abelian groups, 2×1 is not equal to 1×2). The A field is referred to as a potential gauge field of SU-2 symmetry. Present day electrical engineering books either omit or just briefly mention the existence of the A fields. Heaviside interpretation of

Maxwell's equations is used as a mathematical convenience in present electrical engineering courses. In order to solve Maxwell's equations without losing the information contained in the A fields, CLIFFORD algebra, which includes parts of Hamilton's quaternion theory, is necessary. The entire field of quantum physics is based on the physical significance of the A field. No longer is it possible to talk in terms of 3 dimensions - 4 dimensions are necessary to describe the A fields. This is because the matrices describing the Stokes parameters are 2×2 ; there are two of these matrices which means 4 dimensions. All the transients involved are SU-2 symmetry. Both the amplitude and the frequencies do vary. A mathematical way to visualize the amplitude and frequency variations is to use a Poincaré sphere with the amplitude changing rotation along the latitudes and polarization changing along the longitudes. Solitons (a non-linear wave traveling through a medium) are also SU-2 symmetry. In SU-2 form, charge does not repel (see Gilmore's book on 'Group Theory' for further information). An interesting point is that magnetic charge does not exist in SU-2 symmetry and when SU-2 symmetry is broken, the magnetic charge doesn't come with it. The transformations necessary in CLIFFORD algebra are referred to as Loxodromic transformations. This basically means how do you convert geometric images pertaining to scaling on a normal course into algebraic expressions. Quaternions are the mathematics of SU-2 group, but quaternions conserve everything and never seem to converge. CLIFFORD's algebra is best and leads to the more familiar Smith charts with which electrical engineers are familiar. The original Tesla coils, for instance, were able to generate fields of SU-2 symmetry. However, the coils described in Tesla's papers do not generate A fields of SU-2 symmetry as do those he personally built. We are attempting to obtain an original Tesla wound coil which we will x-ray to see if the windings are counter wound. Counter winding the coil windings would allow A fields to be generated.

VERIFICATION OF "A" VECTOR FIELDS. The "A" Vector field is referred to as a curl free vector potential. A Torroidal coil which is shielded to reduce any "B" field interference has been designed. Stainless steel wave guides are used in conjunction with this torroidal coil and its detector to cut down on heat transfer, both from and to liquid helium. A Josephson detector is used to detect the "A" field interaction with scalar fields. Because the "A" vector is a scalar wave potential, the common electromagnetic antenna may not be used. The Soviet literature describes such a device and apparently is in every day use in their communications. The "A" field effects only phase shifts. Because of this, an "A" field vector can get through metals and could be used as a bomb detection device. Another opportunity would be to design non-nuclear rods for existing nuclear power stations to generate heat using "A" fields. A device is also under development to sterilize and remove the dead bacteria from medical instruments. Viruses are also destroyed by the "A" fields. Heating, however, does not destroy viruses but only bacteria. As an aside, the topic of the AIDS virus came up and the individual doing this work indicated the pore size of the rubber used to manufacture condoms is large enough to allow the AIDS virus to flow through. A brief discussion ensued as to how you could use the "A" field in conjunction with a condom to destroy the AIDS virus. A quick coffee break followed.

SOVIET APPLICATIONS OF TESLA TECHNOLOGY TO ZERO POINT ENERGY. The Kraysnoiarsk radar is not a radar. When Secretary of Defense Chaney took a tour through this facility, he was always looking for radar scopes which are nonexistent. The Russians told him it was a scalar wave transmitter to transfer SU-2 power to a satellite and then to Moscow. Because we have no such knowledge of devices such as this in our country, we continue to talk as though this transmitter is a radar. We all believe the Russians will probably dismantle this facility and figure out a way to redesign the outward appearance to obviate our concerns. We have experience in lasers (light transmission); the Soviets use electromagnetic radiation and transmitted electromagnetic radiation in most of their studies. Tesla, for instance, was convinced he had tapped the energy of the ether (cosmos) at Colorado Springs. The Soviets have been doing work on beta decay and have found beta decay slows by 6 percent due to the existence and presence of "A" fields. The Soviets are extremely interested in parapsychology and have sisters which do a teleportation display. When they do this display in the presence of a self-discharge tube, the tube dims 80 percent. The Russians feel man was able to tap zero potential energy fields for thought. Man, as a society, has lost the ability to tap zero potential energy for thought over the eons. The Soviets also use optical computers to monitor SU-2 symmetries. They have determined what energy levels are emitted from various organs in a person's body, and can monitor these various energy levels using optical computers. Before treating anyone, the person simply lies in a normal hospital bed equipped with the optical sensors and his body is scanned. From the output of the optical computer, the Soviets know the specific organ and the treatment. The greatest problem the Russians are faced with is they cannot make more than a few devices and need our technology to mass produce items. They would like to give the United States their technology on scalar waves and zero point energy in return for our technology on how to produce and manufacture goods. The Soviets also presently use plasma jet technology for microwave energy transmission. Both beamed and reflective power systems are employed in the Soviet Union. A major problem for both the Soviets and us in interpreting Tesla's writings is that much of it is hand written in 5 different languages within the same document. Another interesting fact is that Poincare was ostracized by the West in the 1850's. A Soviet mathematician, Lipanov, believed in him and because of this belief, Soviet mathematics is extremely powerful. A for instance was given regarding the solution of complex vector equations which we need a CRAY computer to solve. The Soviets can solve problems of this magnitude by hand calculators.

RUSSIAN - JAPANESE SPACE LAUNCH AGREEMENT. As our meeting was coming to a close, mention was made of an agreement signed between Japan and the Soviet Union regarding space. The Soviets will transfer to Japan all their rocket technology and Japan in turn will supply all technology for the moon base, a joint Japanese Soviet space station, and a joint Mars program. The Japanese and Soviets have also agreed to share in developing space manufacturing. The general goal of the Soviets is to tie all of Europe together.

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Russia gets the manufacturing technology and everyone else gets high technology developed by the Soviets. What is needed to establish space bases is energy. The ability to tap into zero point energy opens the door for tremendous advances in space manufacturing. It also offers an environmentally benign source of terrestrial power.

FRENCH COMMERCIAL SATELLITE LAUNCHES. The DIA, supported by others in the room, indicated the French have tied up 75 commercial satellite launches. There remain an additional 25 commercial launches and the probability is the French will receive contracts to launch a majority of these 25. Should the French tie up 100 commercial launches, it leaves us 15 launches for the military. It takes about 9 months for all the clearances to go through in order to launch a commercial satellite from the Cape or Vandenberg. Our technology at the launch complexes is at least 30 years old. And yet we protect it like we've just discovered it yesterday. We are so security conscious it just takes too long for people to use our launch vehicles. Hence, the success of the French, or at least this is the reason given for the French success by DIA.

RUSSIAN TRIP. The Soviets are interested in having a group of western scientists come to Novosibirsk and verify their results on zero point energy. The DIA feels the Soviets may be willing to pay for us to visit and verify their experiments on zero point energy. The trip is tentatively planned for March 1990. Should the Soviets elect not to support us, the DIA has indicated their willingness to send us. No longer are the Soviets closing their facilities to U.S. people. Dr. S.N. B. Murthy/Purdue University was one of the first westerners to be invited by the Soviets to see their hypersonic engine test facilities at the Moscow Aviation Institute outside of Moscow, Russia. This facility was considered top secret, and off limits to any westerner prior to Glasnost. Whether the Russian trip happens or not, zero point energy is a reality and it's application to today's problems unique. Harnessing zero point energy for propulsion, while not considered in this meeting, may be possible through the use of either the Biefeld-Brown or Casimier effects.